
HOMEWORK DAY 12 – *Trigonometric integrals §7.2 (examples 1-6,9)*

1. Evaluate the following integrals.

(a) $\int \sin^3 x \cos^2 x \, dx$

(b) $\int_0^{\pi/2} \cos^9 x \sin^5 x \, dx$

(c) $\int_0^{\pi/2} \cos^2 t \, dt$

(d) $\int \tan x \sec^3 x \, dx$

(e) $\int_0^{\pi/2} \cos^5(t) \, dt$

(f) $\int_0^{\pi/4} \sin^2(2\theta) \, d\theta$

(g) $\int \tan^2 x \, dx$

(h) $\int \frac{\cos^5 \alpha}{\sqrt{\sin \alpha}} \, d\alpha$

2. (a) Evaluate $\int \sin(3x) \cos(5x) \, dx$ using the appropriate identities in §7.2, Formula 2.

(b) Evaluate $\int \sin(3x) \cos(5x) dx$ using integration by parts.

(c) Evaluate $\int_0^{2\pi} \sin(3x) \cos(5x) dx$.

HOMEWORK DAY 13 – *Trigonometric substitution §7.3*

Evaluate the following integrals, using any method of your choice.

3. $\int \frac{x^3}{\sqrt{1-x^2}} dx$

4. $\int \frac{x^3}{4+x^2} dx$

5. $\int_0^{1/2} x\sqrt{1-4x^2} \, dx$

6. $\int_2^3 \frac{dx}{(x^2 - 1)^{3/2}}$

$$7. \int_0^2 \frac{dt}{\sqrt{4+t^2}}$$

HOMEWORK DAY 14 – *Partial Fractions – linear factors §7.4*

8. §7.4: 1a

9. §7.4: 7

10. §7.4: 11

11. §7.4: 15

12. §7.4: 16

13. §7.4: 18

15. §7.4: 51